

Nate Cirino

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Graduating May 2027 | Available to start May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Foundations of Data Science, Algorithms & Data Structures, Computer Systems, Digital Design & Computer Architecture, Object-Oriented Design, Foundations of Software Engineering, Computer Networks, Systems Security

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack for a large-scale consumer e-commerce retailer across all Wayfair physical brands (AllModern, Birch Lane, Joss & Main, Perigold): e-labels, mobile, and customer-facing POS
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync
- Building a supplier-agnostic e-label API unifying 4 suppliers behind one interface, powering new outlet launches

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Optimized Expert AI retrieval, search, and filtering across 20+ microservices, improving query relevance at scale
- Integrated Expert AI (LLM-backed) into CME accrual/redemption microservices, serving GenAI CDS to 250K+ clinicians
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing inference workflow
- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Flask backend with RAG retrieval pipeline powering personalized workout recommendations via Bedrock KB
- Evolved retrieval/ranking stack from LangChain to Vespa to zero-ops Bedrock KB, lifting relevance and cutting overhead
- Sustained 99.99% uptime across 1–2M monthly model inference calls for 2K paying users via AWS WAF, Datadog, and K8s
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Continue as technical advisor on architecture and agentic-workflow direction

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

NOAA AUV Sonar Detection | *Python, PyTorch, Deep Learning, AWS Neuron SDK, SageMaker* April 2025 – July 2025

- Trained CNN models in PyTorch on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address long-tail class imbalance and sensor noise, improving detection
- Designed AWS Neuron SDK deployment path optimizing onboard accelerator inference under hard real-time limits

Sentinel | *Next.js, FastAPI, Claude, LLM Agents, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an LLM agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG retrieval over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

SKILLS & HONORS

Languages/Tools: Python, C++, Rust, C, Java, TypeScript, JavaScript, Bash, Linux, AWS, Azure, Docker, Kubernetes, Terraform, Jenkins, Buildkite, Groovy, PostgreSQL, GraphQL

ML / Frameworks & Libraries: PyTorch, TensorFlow, Pandas, RAG & vector retrieval (pgvector, Vespa, Bedrock KB), FastAPI, Flask, React, Next.js, Node.js, Express

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor